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Normal Matrix Theorem

A matrix $A \in \mathbb{C}^{n \times n}$ is unitarily diagonalizable iff A is normal.

Principal Axis Theorem

Complex matrix A is Hermitian iff A can be unitarily diagonalized as $A = UDU^H$.

Diagonalization of Quadratic Forms

We can diagonalize a form $q(r_1, ..., r_n)$ and through a change of coordinates of the form $\mathbf{r} = C\mathbf{s}$, construct a new form $q'(s_1, ..., s_n) = \sum a_i s_i^2$.

Equivalent Conditions for Positive Definite Form

Let A be a real symmetric matrix.

(i) eigenvalues of A are positive $\implies$ (ii) A is positive definite

$A = QDQ^T$ with all eigenvalues from $D = \text{diag}(\lambda_1, ..., \lambda_n)$ positive.

Since Q is an orthogonal basis, $\mathbf{x} = x_1\mathbf{q}_1 + \cdots + x_n\mathbf{q}_n$, so $\mathbf{x}^T A\mathbf{x} = \sum \lambda_i x_i^2 > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

(ii) A is positive definite $\implies$ (iii) upper left $k \times k$ submatrix A_k has $\det(A_k) > 0 \quad \forall k$

We can view vectors in $\mathbb{R}^k$ as vectors in $\mathbb{R}^n$ but with the upper values set to 0

$$\mathbb{R}^k = \{\mathbf{x} \in \mathbb{R}^n | x_i = 0 \quad \forall i > k\}$$

Since A is positive definite, so is A_k, which means all eigenvalues are positive, and so $\det(A_k) > 0$.

(iii) $\det(A_k) > 0 \quad \forall k \implies$ **(iv)** $A = LDL^T$ **with all** $d_i > 0$ **for** $D = \text{diag}(d_1, ..., d_n)$

Immediate from prop 4.17: *A can be written in the form LDU iff A_k is invertible for all $k = 1, ..., n$.*

(iv) $A = LDL^T$ **with all** $d_i > 0 \implies$ **(i) eigenvalues of** A **are positive**

From Sylvester's law of inertia.

Positive Semidefinite Forms

Real Symmetric Matrix is Positive Semidefinite iff $A = LDL^T$ with D Nonnegative

A can be decomposed as $LPDL^T$ where L is orthogonal, and since $A = A^T$, so PD is symmetric. Then we see $P = I$, and D entries are nonnegative.

Sylvester's Law of Inertia

Let A and B be congruent real symmetric matrices. Then they have the same signature- number of positive, zero and negative eigenvalues. Likewise real symmetric matrices with the same signature are congruent.

- We easily see $n_0(A) = n_0(B)$ since they share the same null space.
- When diagonalizing real symmetric matrices, there's multiple ways to order the diagonal, so we reorder it so the positive ones are first.
- Now we only have to look at D, E diagonal matrices since A and B are congruent to them.
- We assume $n_+(D) < n_+(E)$
- $f_j(x_1, ..., x_t)$ is a linear function in t variables. For $j = 1, ..., s$ we can pick $\mathbf{a} = \begin{pmatrix} a_1 & \cdots & a_t \end{pmatrix}$ so that all $f_j(\mathbf{a}) = 0$.
    - To see this, simply make $F \in \mathbb{R}^{s \times t}$ be the matrix with the coeffs of $f_1, ..., f_s$ as its rows and $\mathbf{a} \in \mathbb{R}^t$. We see the null space of F is nonzero since $s < t$.

Exercises

9.1.6

Since iA is Hermitian and only has real eigenvalues, then A has only imaginary eigenvalues. Write $\det(A) = i^n \lambda_1 \cdots \lambda_n$ where λ_i are the eigenvalues of iA.

By this result, if A is skew-symmetric, then $\det(S) = 0$ since $\det(S)$ is pure imaginary.

For n even, the eigenvalues occur in pairs $\pm i\lambda_k$, which means $\det(S) \geq 0$.

9.1.7

Write A in unitary diagonal form, then note $D = \text{diag}(\lambda_1, ..., \lambda_n)$. Since $\det(A) = \det(D)$ and the eigenvalues of a Hermitian matrix are real so $\det(A) \in \mathbb{R}$.

9.1.8

Follows immediately from $NU = UD$, where D are the eigenvalues, and $U = \begin{pmatrix} \mathbf{u}_1 & ... & \mathbf{u}_n \end{pmatrix}$ are the eigenvectors. Since U is unitary we have $UU^H = I$ which means the eigenvectors are all Hermitian orthonormal.

9.1.9

See wiki for such a matrix.

9.1.10

$UU^H = I \implies \mathbf{u}_i \bar{\mathbf{u}}_i = N(\mathbf{u}_i) = 1$, but $|\mathbf{u}_i| = \sqrt{N(\mathbf{u}_i)} \implies |\mathbf{u}_i| = 1$.

$$|\det(U)| = \sqrt{N(\lambda_1 \cdots \lambda_n)} = \sqrt{N(\lambda_1) \cdots N(\lambda_n)} = 1$$

Let $x = a_1 + ib_1, y = a_2 + ib_2$, then let $N(x) = N(y) = 1$. We see

$$N(x+y) = (a_1+b_1)^2 + (a_2+b_2)^2 \leq a_1^2 + b_1^2 + a_2^2 + b_2^2 = N(x) + N(y)$$

$$\implies |Tr(U)| \leq n$$

9.1.11

$QQ^T = I \implies \det(Q) = \pm 1$. But $\det(Q) = \lambda_1 \cdots \lambda_n$ with all λ_i imaginary. Therefore the eigenvalues are in conjugate pairs $\pm i\mu_k$ and n is even.

9.1.14/15

Needs Fuglede's theorem since $AB = BA$ and $A^H B^H = B^H A^H$ does not imply that A, B, A^H, B^H all commute. Given that and the earlier result on diagonalizable matrices, as well as $U^{-1} = U^H$, the questions are easy to solve.

For the next one, just note $A + B = U(D_1 + D_2)U^H$ means the eigenvalues are $D_1 + D_2$.

9.1.17

(iii) $AA^T = UD_1U^T = (A^TA)^T = (UD_2U^T)^T \implies D_1 = D_2$.

9.2.10 (vi)

Using $B = CAC^T$ and $A = A^T$, we get $AB = ACAC^T = BA = CAC^TA \implies AB = (AB)^T$. So AB must be symmetric, which is not always true.

9.2.11

$A = QDQ^T$ with $D = \text{diag}\{d_1, ..., d_n\}$ all nonnegative. Therefore we can take any arbitrary kth root K in $\mathbb{R}^{n\times n}$. Let $M = QKQ^T$, then we see $A = M^k$.

9.2.12

Let $U \subset V$ be the subspace spanned by B. Since A is symmetric diagonalizable, it is a semisimple linear mapping. Since it is positive definite with only positive eigenvalues, therefore the span of A is $V \implies \text{span}\, B \subset \text{span}\, A$.

Let $\{\mathbf{v}_1, ..., \mathbf{v}_n\}$ be an eigenbasis for A. When $B\mathbf{v}_1 = \mathbf{0}$, then $\mathbf{v}_2^T AB\mathbf{v}_1 = \mathbf{0}$. But $(\mathbf{v}_2^T AB\mathbf{v}_1)^T = \lambda \mathbf{v}_1^T B\mathbf{v}_2$ where λ is an eigenvalue of $B\mathbf{v}_2$. Since B is symmetric, we have $\mathbf{v}_1^T B^T = \mathbf{v}_1^T B = \mathbf{0} \implies \mathbf{v}_1^T AB\mathbf{v}_2 = \mathbf{0}$.

Now assume $B\mathbf{v}_1, B\mathbf{v}_2 \neq 0$. Let λ be the eigenvalue of $B\mathbf{v}_2$, then we see that $\mathbf{v}_1^T AB\mathbf{v}_2 = \lambda \mathbf{v}_1^T B\mathbf{v}_2 = \lambda(\mathbf{v}_1^T B\mathbf{v}_2)^T = \lambda \mathbf{v}_2^T B\mathbf{v}_1 > 0$.

We conclude that $QABQ^{-1}$ is symmetric where $Q = \begin{pmatrix} \mathbf{v}_1 & ... & \mathbf{v}_n \end{pmatrix}$.